ME2110 — Section A11

Final Report

How to Train Your Robot (Team 6):

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Abstract

Chief Hiccup needs an autonomous robot to help complete tasks around his village of Berk. The most difficult of these tasks is refilling the dragon feeding stations and lighting the bonfire. Important engineering specifications to carry out these tasks include the robot being deactivated and static before and after the 40-second operation period, being able to operate fully autonomously for the 40-second operation period, and a minimum linear motion of 0.0625 in/s in the forward direction. A potential solution for the critical tasks comes from a diameter-based diverter that sorts fish and eels onto curved transportation slides, and scissor links powered through pneumatics that extend and contract to collect the torches. The final design incorporates a rack and pinion torch mechanism, scissor link water lever activation, piston powered sheep pusher, and foldable slides for the fish and eel task. The total Bill of Materials for this design costs $103.70 with fabrication consisting of laser cutting and 3D printing to create a total of 499 components. Of 3 alternative designs, the final design most resembles Hiccup, which scores the highest with 350, 80.13% of the total possible points. The final robot received an average design review score of 3.52 out of 4 and scored a cumulative total of 180 points during the final competition. The robot has a modular design, but field tolerances and alignment issues limited its competition score, indicating that additional testing would significantly improve consistency in scoring.

Introduction

Chief Hiccup needs the engineers of Berk to build an autonomous robot to carry out his village tasks. Tasks include leaving the dragon stables, refilling the dragon feeding stations, returning loose sheep to their paddocks, lighting the bonfire with torches, and extinguishing the fire using a water bucket positioned 42” above the ground. The robot must also complete these tasks within 40 seconds, and compete against 3 other robots. The engineering team has 8 weeks to design, fabricate, and test the robot with a maximum budget of $120 for the final Bill of Materials (BOM). Design challenges stem from limited time and budget, forcing tradeoffs in cheaper time-consuming manual fabrication versus high-cost rapid prototyping.

Problem Understanding

Chief Hiccup needs the robot to perform consistently, so it must be robust and repairable. The robot must also be safe for the environment and villagers. While competing, it must be inactive before and after the operation period, and comply with the Competition Rules [1]. Aside from robot activation, which is needed to complete any other task, the robot should prioritize based on point values, full water lever activation, and alignment of both torches on the bonfire center, while avoiding multiple sheep in one paddock or eels in the feeding trough.

Table 1 shows the engineering specification sheet to fulfill Chief Hiccup’s needs. General specifications require a $120 BOM limit, safe exposed robot parts, and a 3-minute time limit to prepare the robot for each competition round. Physical requirements constrain the robot to a minimum volume of 60 in^3 to house 4 sheep, maximum dimensions of 12”x24”x18”, and vertical extension: >42” from the ground to hit the water lever. Electrical specifications state that two 120V AC outlets will supply the robot with power, and the Arduino and control board use input from banana plugs. Track dimensions and maximum operation time generate the following performance parameters: linear motion forward for leaving dragon stables: >0.0625 inches/sec, linear motion left and right for the light bonfire task: >0.525 inches/sec, and linear motion upward for the extinguish fire task: >0.9 inches/sec. The robot must also be inactive before and after the operation and run fully autonomously for under 40 seconds.

From the House of Quality (HOQ) in Figure 1, Chief Hiccup’s highest weighted robot requirements include: inactive before and after each round, complies with competition rules, successfully deploys, and fully leaves dragon stables. With a technical importance rating of 219 (12% of the relative weight), the most important engineering requirement is the robot being deactivated and static before and after the operation period. Other key engineering requirements are autonomous operation of the robot: <40 seconds and linear motion forward: >0.0625 inches/sec. The HOQ also indicates a negative correlation and tradeoff between the minimum linear motion forward and upward, as reaching the great hall faster decreases the distance of the robot to the lever. A positive correlation exists between the robot being deactivated and static before and after the operation period, and the Arduino and control board receiving input from the banana plugs. This synergy occurs because connecting the banana plugs to the robot activates it.

Conceptual Design

To complete all the tasks while adhering to the Competition Rules, the physical mechanisms and solutions needed for the six primary functions are listed in Figure 2. The robot must deploy with banana plugs and leave the dragon station to activate the system. The Morphological Chart in Figure 3 depicts motor-powered movement of the robot in any direction as a solution to this function. The robot also has to extinguish the fire by positioning underneath the water bucket lever and activating the lever. Hard-coding, sensors, or a bump stop can align the robot, while pneumatics or a motor winch mechanism can extend drawer slides or scissorlinks to the lever. Refilling the dragon feeding stations requires collecting, sorting, and delivering fish and eels to their collection zone. A straight or curved arm superjacent to the centerpiece that feeds into a physical filter based on fish and eel dimensions solves the collection and sorting process, and connecting slides or tubes can be used to transport the animals. The robot must also hold 4 loose sheep and return them to their unique paddock. Sheep may be held by vertical or horizontal stacking in a sheep container. Sheep deployment methods include gravity-powered slides, a spring-activated pusher, or a vertical lever. Lighting the bonfire consists of moving torches located to the left and right of the bonfire as close to the bonfire's center as possible. Pneumatic powered levers or scissorlinks, or motor powered rack and pinions are possible torch movement mechanisms. Torch alignment may be done through an IR sensor, hard-coding the movement mechanisms, or an alignment notch underneath the robot. After the 40-second operation period ends, the robot must deactivate and become static, which can be completed through removing power to the robot or a timer built into the code.

Design Overview

The desired design was selected from the yellow-highlighted boxes listed in the Morphological Chart in Figure 3 by testing and fabricating multiple designs from previous

Sprints. Figure 4 shows the full CAD assembly of the robot, including a drive base, a sheep and water lever subsystem, a torch subsystem, and a fish and eel subsystem. The robot measures 11.7”x21.3”x17.7”, meeting the boxing specification of 12”x24”x18”. As shown in Figure 5, the robot expands to 31.6” wide and 26.3” long with the fish and eel slides unfolded.

To leave the dragon stables, this design consists of a motor-driven drive base with three 3D-printed wheels wrapped in a rubber strip as outlined in Figure 6. Three wheels allow the robot to always maintain full contact with the ground. The drive gear shown is mounted to the motor axle and meshes with a gear on the front wheel with a gear ratio of 3 to 1, moving the robot quickly to the Great Hall in approximately 1.5 seconds, potentially increasing fish and eel collection. All wheels have press-fitted flange bearings to minimize friction.

Figure 7 shows the expanded fish and eel slides, containing a collector arm, sorter, and separating slides. The collector arm is placed above the Great Hall and is angled to direct all fish and eels into the sorter, which has a height-based filter that passes fish while blocking eels, rerouting them to the left-hand ramp. Before falling into the slides, each fish and eel falls on their side, so fish and eels roll vertically sideways down the slides. This feature allows the slides to stay horizontally compact while efficiently deploying fish and eels to their respective zones.

The torch mechanism in Figure 8 is a rack-and-pinion design, in which a motor drives the pinion gear and two attached racks move in opposite directions. Each rack ends with a torch striker, which pushes the torches into the bonfire. During expansion, shown in Figure 9, the torch strikers are 23.6” apart, providing ~0.5” clearance from each torch. Each torch striker has a protrusion that holds the fish and eel slides during boxing. At activation, the motor drives the pinion slightly, moving the torch strikers outward to free the slides from the containment blocks, causing expansion. Figure 10 shows this behavior on the real robot boxed configuration. The rack and pinion housing allows fastening to the drive base.

Figure 11 shows the sheep and water lever mechanisms subassembly. The sheep are stacked vertically in the sheep container; as the bottom sheep is removed, the rest slide down so the next sheep is next to be fired. Figure12 shows the robot’s front view, revealing exit hole for the sheep from the front of the 3D-printed sheep container. Figure 13 portrays the sheep striker attached to the sheep pneumatic piston so that the two move in unison. When actuated, the sheep piston pushes the bottommost sheep into the sheep paddock, then retracts, allowing the next sheep to fall. The water lever subassembly in Figure 14 is made of a piston and vertically contracting and expanding scissor links. The side view in Figure 15 shows the scissor links are made of 3D printed PLA for more impact resistance and toughness, while being fastened to a vertical ¼" MDF board. Figure 16 displays full expansion of the scissor links to 35.6", which reaches the water lever height of 42" due to a ground offset of 10".

The algorithm outlined in Figure 17 illustrates that the robot activates when the banana plugs are connected to meet the engineering specification, including a debouncing technique to mitigate noise. The rack and pinion mechanism expands slightly after activation to unfold the fish and eel slides, enabling the robot to travel to the Great Hall and start collecting fish and eels. The algorithm fires the water lever piston first as it requires the most force. Sheep deployment uses IR sensor peak detection as paddock walls pass; after a peak, a sheep deploys with a timed offset. At the end, the robot moves backwards while expanding the torch mechanism further, then drives forward so the torches are inside the torch strikers, and collapses the rack and pinion to drag the torches into the bonfire with the torch strikers; all of these operations are time-based.

The total cost of our robot fabrication, as shown in Figure 18 ended up being $103.70, which falls within the BOM requirement of a total cost of $120. All of our fabricated parts and purchased parts are shown in Figure 19.

Alternative Designs

The Hiccup design uses a motor to drive and activate the system. To extinguish the fire, a pneumatic piston powers scissor links to activate the water bucket. Refilling the dragon feeding station is accomplished through a flat collection arm and a height-based diverter that splits the animals onto their slides. Sheep are vertically stacked and dispensed through a motor-powered mechanism. Underneath, a scissor link system collects the torches, and an alignment notch centers them on the bonfire. To deactivate, the robot turns off if power is removed.

The Astrid design uses the same solutions for activating and deactivating the system. A drawer slide and motor-winch perform the extinguishing fire. Fish and eels are collected with a curved flat arm and delivered to their zones through a diameter diverter and curved slide. The sheep are stored horizontally and then deployed into their paddocks through a passive horizontal lever. Finally, a rack and pinion transports and aligns the torches to light the bonfire.

The Fishlegs design also uses the same principles for activating and deactivating the system. Extinguishing the fire uses concentric tubes of decreasing diameters that extend through pneumatic pressure. A right-angle arm collects the fish and eels, and a coin slot diverter sorts them. To deliver the fish and eels, they are moved to the collection zones with a sweeping mechanism. The sheep are stored vertically and deposited into the paddocks by a piston. Lastly, torches are moved onto the bonfire by a sweeping mechanism and aligned through an IR sensor.

Figure 20 shows that the Hiccup design scores the highest out of all three designs in a conceptual evaluation matrix with a total score of 359, which is 80.13% of the total points available. Based on these results, the final design utilizes most elements from Hiccup, and the high-scoring torch mechanism and from Astrid.

Performance Results

The final competition occurred on Friday, November 14, 2025, consisting of a design review by judges and a competition of 72 robots. As shown in Table 2, the design review score was 3.54/4. Judges noted that the system explanation was clear and highlighted the robot's modularity, with subassemblies operating independently. The judges appreciated the evolution of the iterations, as all CAD designs for each Sprint were from scratch. In the competition, the robot scored 97 points on the first round and 83 points on the second, as shown in Table _, neither of which advanced to the eliminations. Activation, leaving the dragon stable, extinguishing fire, and fish and eel slides deployment were consistently effective. The main source of downfall for the robot was the tolerance on the field. One track's Great Hall was especially higher than those on the other tracks, causing the Fish and Eel collector arm to collide with the Great Hall instead of sliding over it, causing the robot to shift completely and lose contact with the Great Hall. As a result, the robot misaligned, causing a failure of sheep deployment in some of the paddocks and torch mechanism alignment. Many fish and eels got stuck and fell right in front of the robot instead of falling into the slides and into their respective collection zones, adding negative points to the score. Ultimately, the robot suffered from field tolerances, which disrupted the flow of many other tasks the robot was aiming for.

Conclusion

The developed robot successfully achieves the five tasks required by Chief Hiccup, with the first design alternative chosen, fabricated, and then improved. While the robot demonstrated strong subsystem functionality and consistent activation, field tolerances and alignment issues ultimately limited overall performance during the competition. With more time allocated to robustness testing in different fields to account for tolerances, the robot could have small alterations to achieve more consistent and higher scores with its consistent subsystems.

Appendix

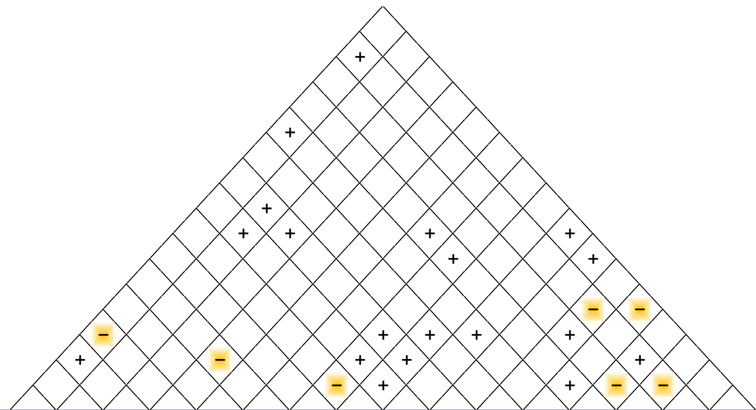
Table 1: Engineering Specification Sheet

				Issued: 9/15/25		
			For: ME2110 Design Report		**Page:**	1
			Specification			
No.	**Date**	**D/W**	**Requirements**	**Responsible**	**Source/Rationale**	**How Validated**
General						
1	9/27/2025	D	Total cost of materials: <$120	Joy	Competition Rules state maximum budget of $120 for building robot.	BOM
2	9/27/2025	D	Exposed materials must be safe (e.g. not damage the track, modifiable to dull sharp edges)	Megan	Competition Rules state that the robot must not stain, permanently change, or damage the competition track and competition items.	Physical Inspection
3	9/27/2025	D	Time to assemble/prepare robot: <3 min	All team members	Competition Rules state that Preparation Time is 3 minutes.	Timer
4	9/27/2025	D	Time to repair robot: >5 min	Carlos	Competition Rules state minumum of 5 minutes in between each round in final competition.	Timer
Physical Characteristics						
5	9/27/2025	D	Robot dimensions when completely collapsed: <12"x24'x18"	Carlos	Competition Rules state that robot must fit within Go/No Go gauge box and stay within those dimensions after the box is removed.	Measuring Caliper
6	9/27/2025	W	Robot volume: >60 in^3	Carlos	Competition Rules state that the robot must hold 4 sheep inside the robot. Each sheep is 3"x2.5"x2" (15 in^3).	Measuring Caliper
7	9/27/2025	W	Robot's vertical extension: >42" from ground	Carlos	Competition Rules state that height of lever on elevated frame is 42" from ground.	Measuring Caliper
8	9/27/2025	W	Weight: <70 lbs	Carlos	Research shows maximum safe lifting weight for woman is 35.274 lb. Assume 2 people carry robot at a time.	Weight Scale
Electrical						
9	9/27/2025	D	Powered supplied by two 120V AC	Nikhil	Competition Rules state the arena is equipped with two 120V AC outlets.	Multimeter
10	9/27/2025	D	ME2110 Arduino and control board receive input from banana plugs.	Nikhil	Competition Rules state the arena is equipped with a set of female connectors that receive the banana plugs.	Visual Check
Mechanical						
11	9/27/2025	D	Pneumatics: <100 psi (7 bar)	Megan	Competition Rules state that any pneumatics must be pressurized to less than or equal to 100 psi.	Visual Check
Performance						
12	9/27/2025	W	Linear motion forward: >0.0625 inches/sec	Joy	Competition Rules state that robot must fully leave dragon stables to complete Activate the System task. Dragon Stable is 2.5" in forward direction. Robot has 40 second operation period.	Speedometer
13	9/27/2025	W	Linear motion upward: >0.9 inches/sec	Joy	Competition Rules state robot must be within 6" of lever height for Extinguish Flame task. Robot has 40 second operation period.	Speedometer
14	9/27/2025	W	Linear motion left and right: >0.525 inches/sec	Joy	Per measuremeents from Competition Rules, robot must reach both torches (placed 10.5" from center of bonfire) and bring torches to center of bonfire. Robot has 40 second operation period.	Speedometer
15	9/27/2025	D	Robot must operate completely autonomously for 40 seconds.	All team members	Competition Rules state that the device must operate completely autonously and is not allowed to utilize or interact with any person or thing during the competition.	Timer
16	9/27/2025	D	Robot is deactivated and static before and after the 40 seconds.	All team members	Competition Rules state no electrical power to actuators, no mechanical motion, and no active release of energy before and after each round.	Visual Check

Correlations	
Positive	+
Negative	−
No Correlation	

Relationships	
Strong	●
Moderate	○
Weak	▽

Direction of Improvement	
Maximize	▲
Target	◇
Minimize	▼



Category	Weight	Customer Requirements (Explicit and Implicit) / Engineering Requirements	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
		Direction of Improvement	▼	◇	◇	◇	▼	▲	▲	▲	◇	◇	◇	◇	▼	▲	▼	▲
		Engineering Requirements	Time to assemble/prepare robot: <3 min	Robot must operate completely autonomously for 40 seconds.	Robot is deactivated and static before and after 40 second operation period.	Time to repair robot: >5 min	Total cost of materials: <$120	Exposed materials must be safe (e.g. not damage the track, modifiable to dull sharp edges)	Linear motion forward: >0.0625 in/s	Linear motion upward: >0.9 in/s	Powered supplied by two 120V AC	ME2110 Arduino and control board receive input from banana plugs.	Robot's vertical extension: >42" from ground	Pneumatics: <100 psi (7 bar)	Weight: <70 lbs	Robot volume: >60 in^3	Robot dimensions when completely collapsed: <12"x24"x18"	Linear motion left and right: >0.525 in/s
Performance	8	Perform consistently	▽			▽												
	9	Able to perform in multiple rounds	○		▽	○												
	10	Inactive before and after each round		○	●													
Properties	7	Easy to maintain and alter	▽			○												
	2	Professional appearance/aesthetics	▽				▽	●										
	10	Comply with competition rules	●	●	●	●	●	●			●	●		●			●	
Tasks	10	Successfully deploy		○	○				○	○								○
	10	Fully leave dragon stables							●						▽			
	9	Activate lever								●			●		▽	▽		
	5	Deliver fish to home zone							▽						▽		▽	
	9	Deliver fish to feeding trough							▽						▽		▽	
	7	Deliver eels to disposal bin							▽						▽		▽	
	9	Deposit each sheep into unique paddock							▽						▽	●	▽	
	7	Move both torches to innermost ring of bonfire												▽	▽		▽	●
		Target	3 min	40 seconds	Visual Check	5 min	$120	Visual check	0.0625 in/s	0.9 in/s	120V AC	Visual Check	42"	100 psi	70 lbs	60 in^3	12"x24"x18"	0.525 in/s
		Max Relationship	9	9	9	9	9	9	9	9	9	9	9	9	1	9	9	9
		Technical Importance Rating	134	150	219	146	92	108	150	111	90	90	81	97	56	90	127	93
		Relative Weight	7%	8%	12%	8%	5%	6%	8%	6%	5%	5%	4%	5%	3%	5%	7%	5%
		Weight Chart																

Figure 1: House of Quality

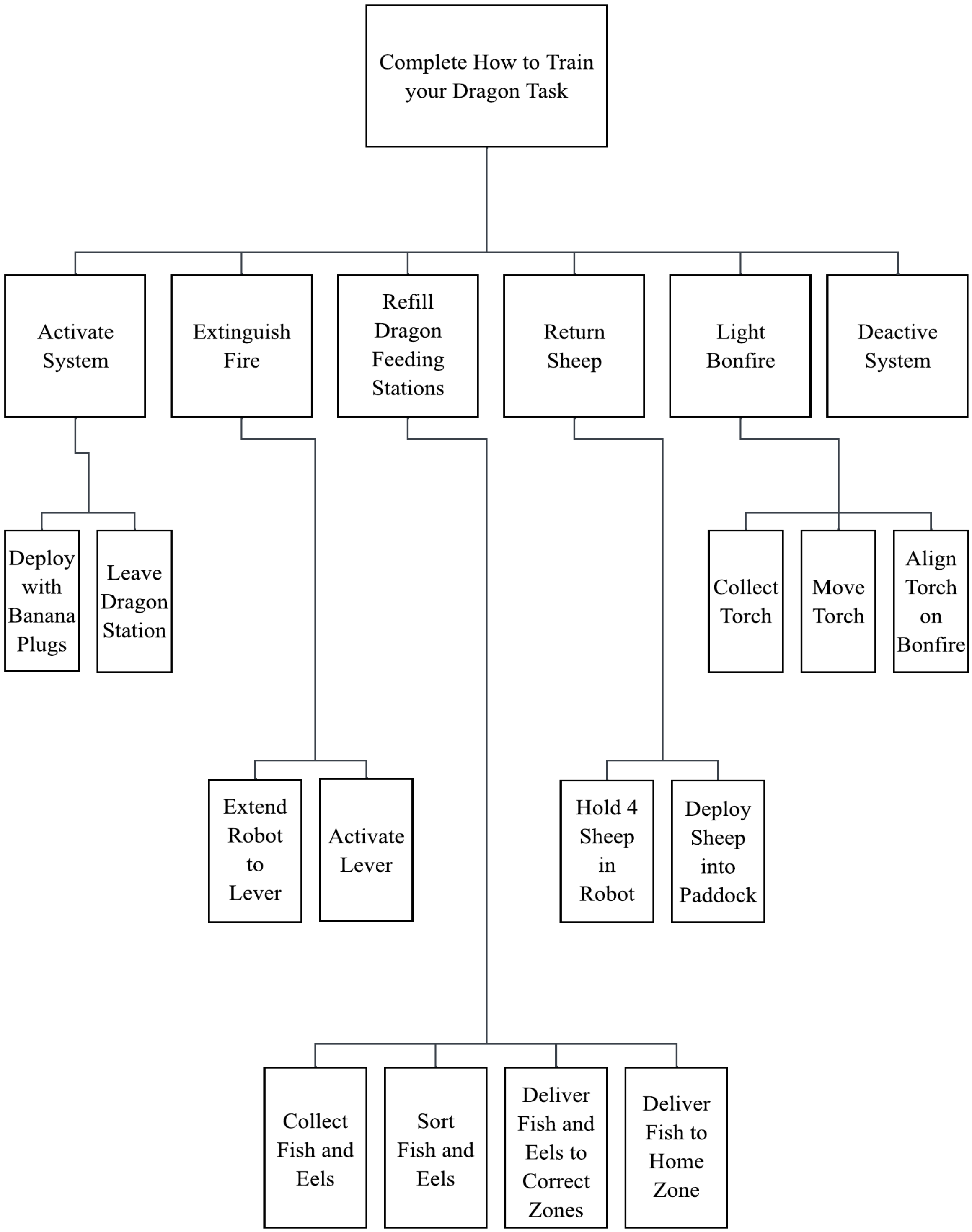
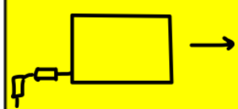
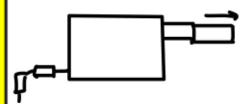
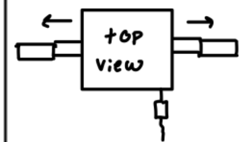
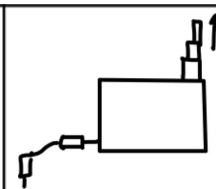
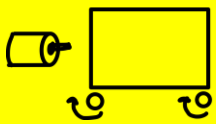
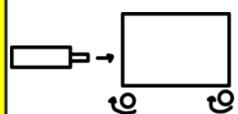
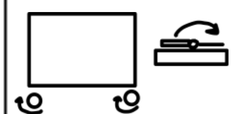
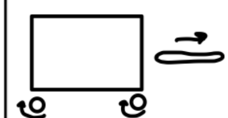
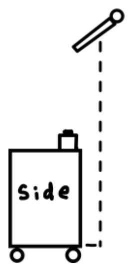
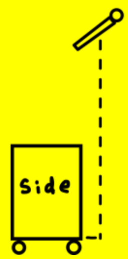
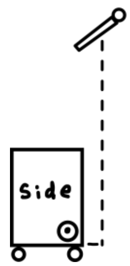
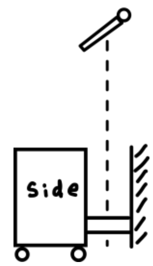
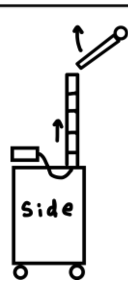
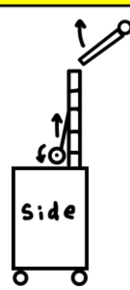
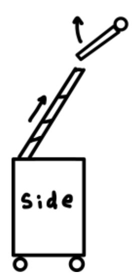
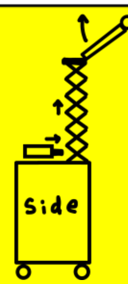
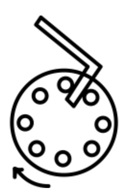
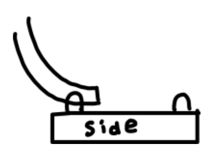
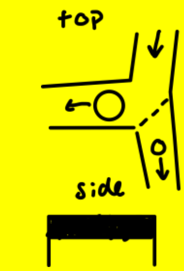
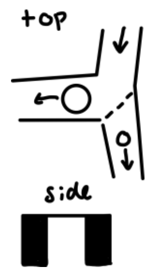
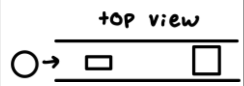
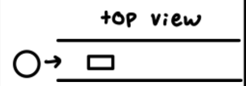
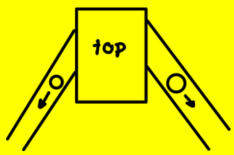
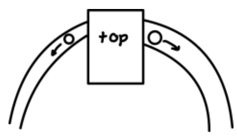
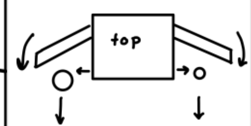
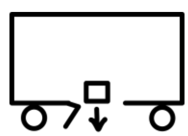
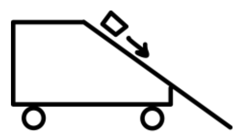
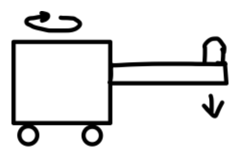
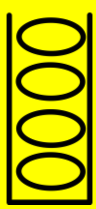
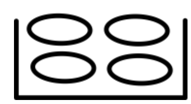
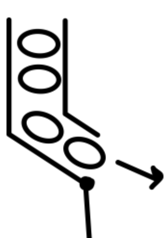
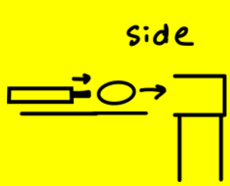
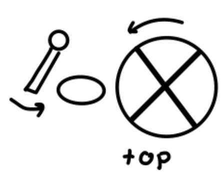
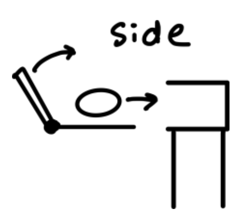
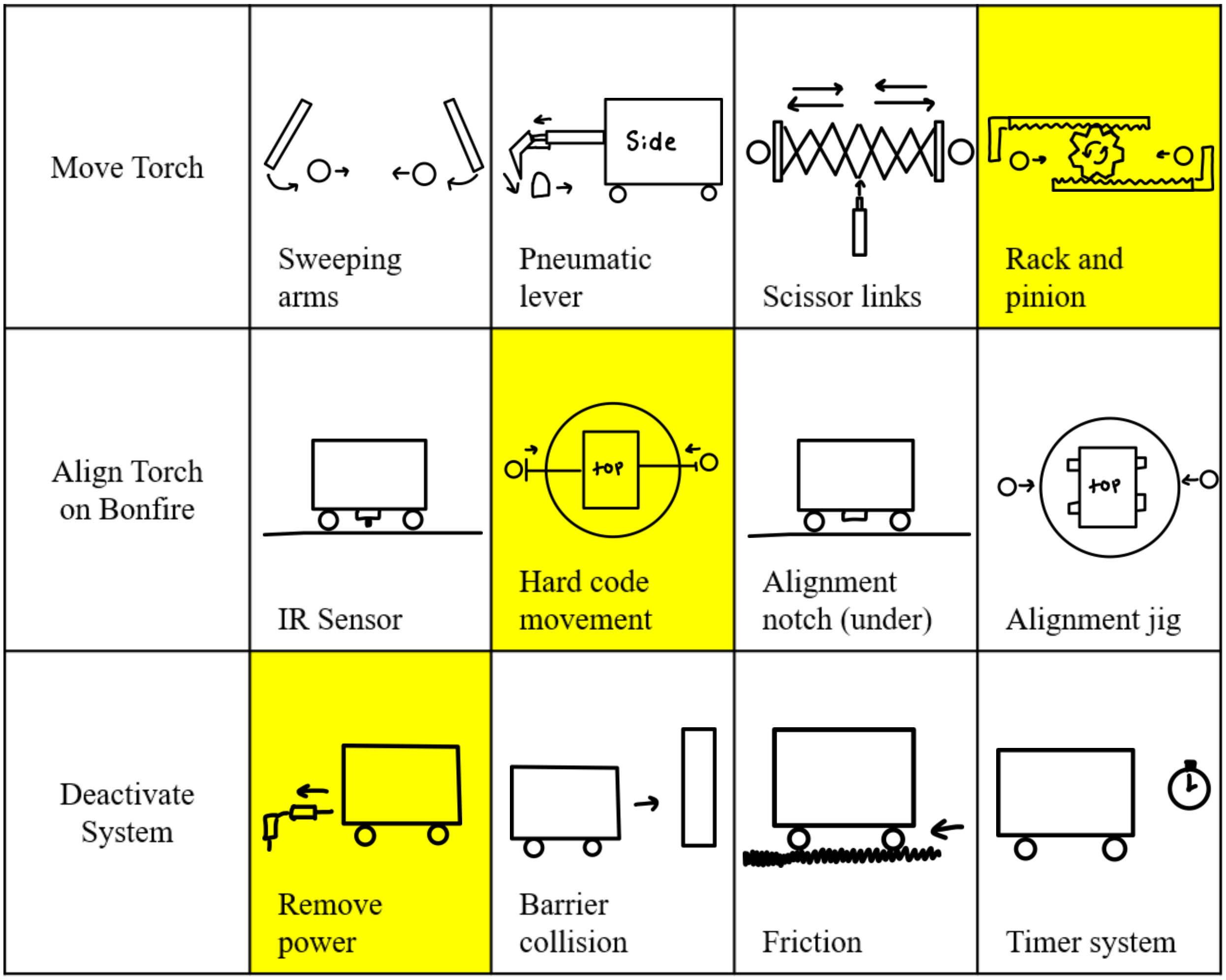


Figure 2: Function Tree

Deploy with Banana Plugs	 Move forward	 Extend forward	 Extend left/right	 Extend upward
Leave Dragon Station	 Motor	 Pneumatics	 Mouse trap	 Rubber band
Position Robot under Lever	 Infrared sensor	 Hard code	 Encoder	 Physical object
Activate Lever	 Pneumatics	 Motor-winch	 Drawer slides	 Scissor links
Collect Fish and Eels	 Flat straight arm	 Flat right-angle arm	 Curved flat arm	 Hook arm

Sort Fish and Eels	 Height diverter	 Diameter diverter	 Coin-slot system	 Coin-slot alternative
Deliver Fish and Eels to Collection Zone	 Straight slide	 Curved slide	 Tube	 Drop and sweep
Deliver Fish to Home Zone	 Chute	 Trap door	 Slide	 Drop and sweep
Hold 4 Sheep in Robot	 Stacked vertically	 Stacked horizontally	 Stored individually	 Stored collectively
Deploy Sheep into Paddock	 Wall drops, sheep roll	 Piston activated pusher	 Passive horizontal lever	 Vertical pushing mechanism



Highlighted boxes denote final design mechanism

Figure 3: Morphological Chart

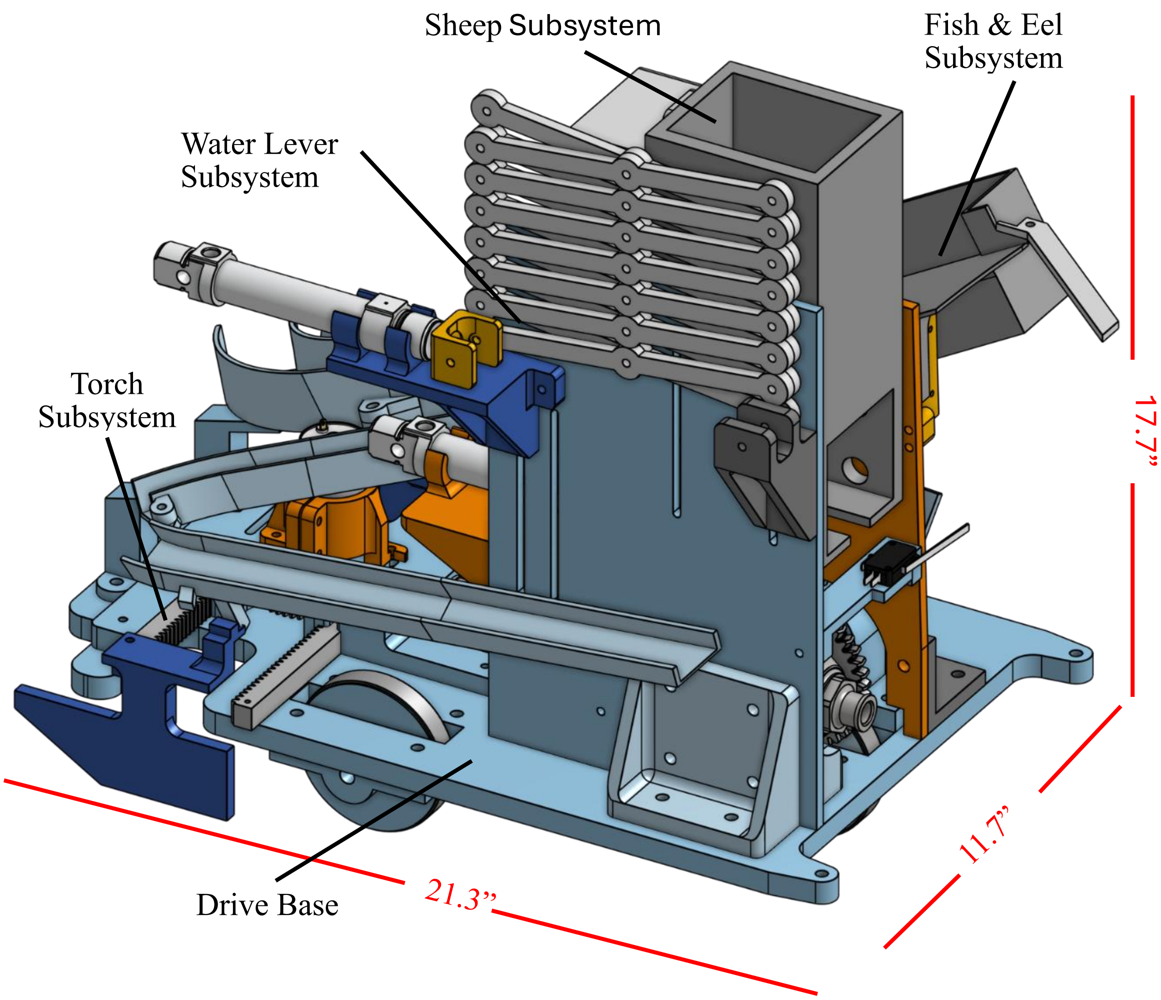


Figure 4: Full Robot

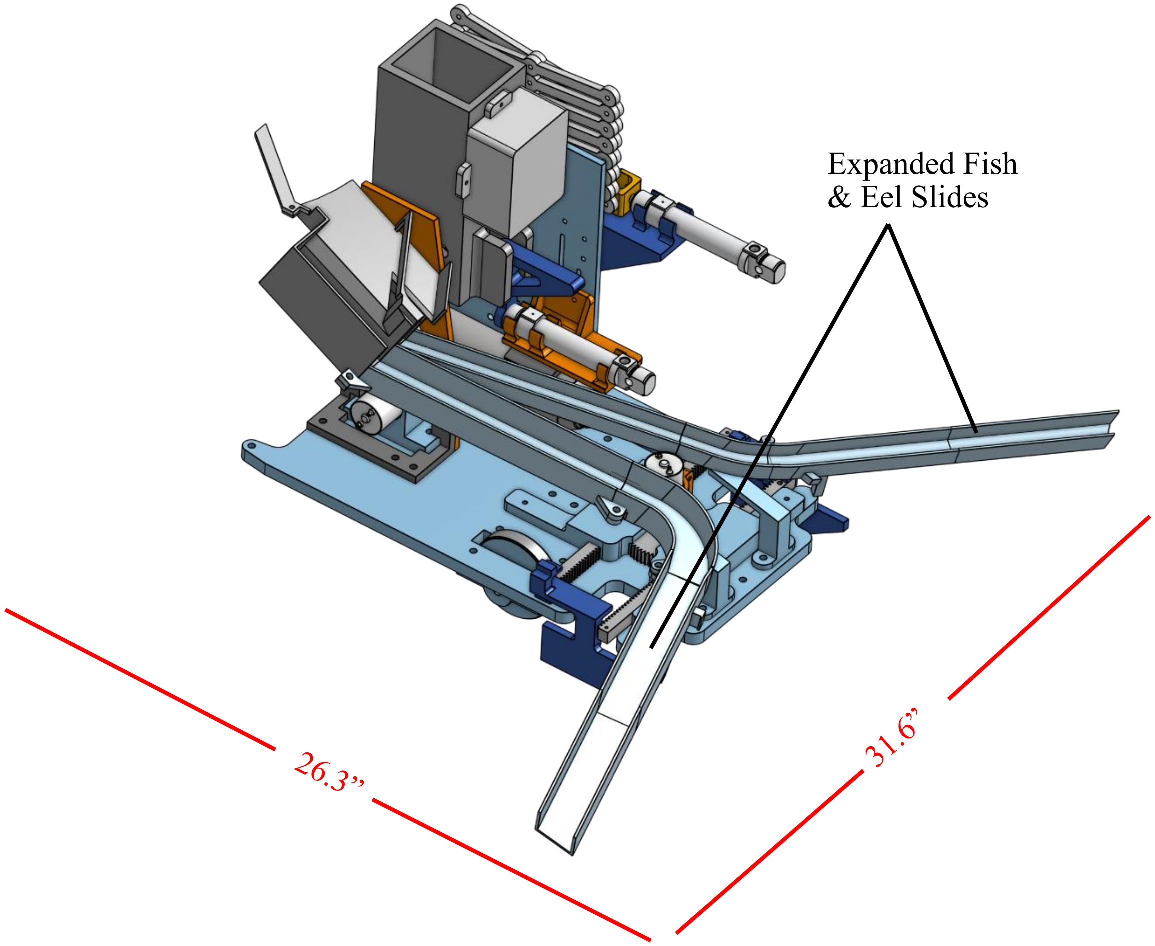


Figure 5: Full Robot with Expanded Fish & Eel Slides

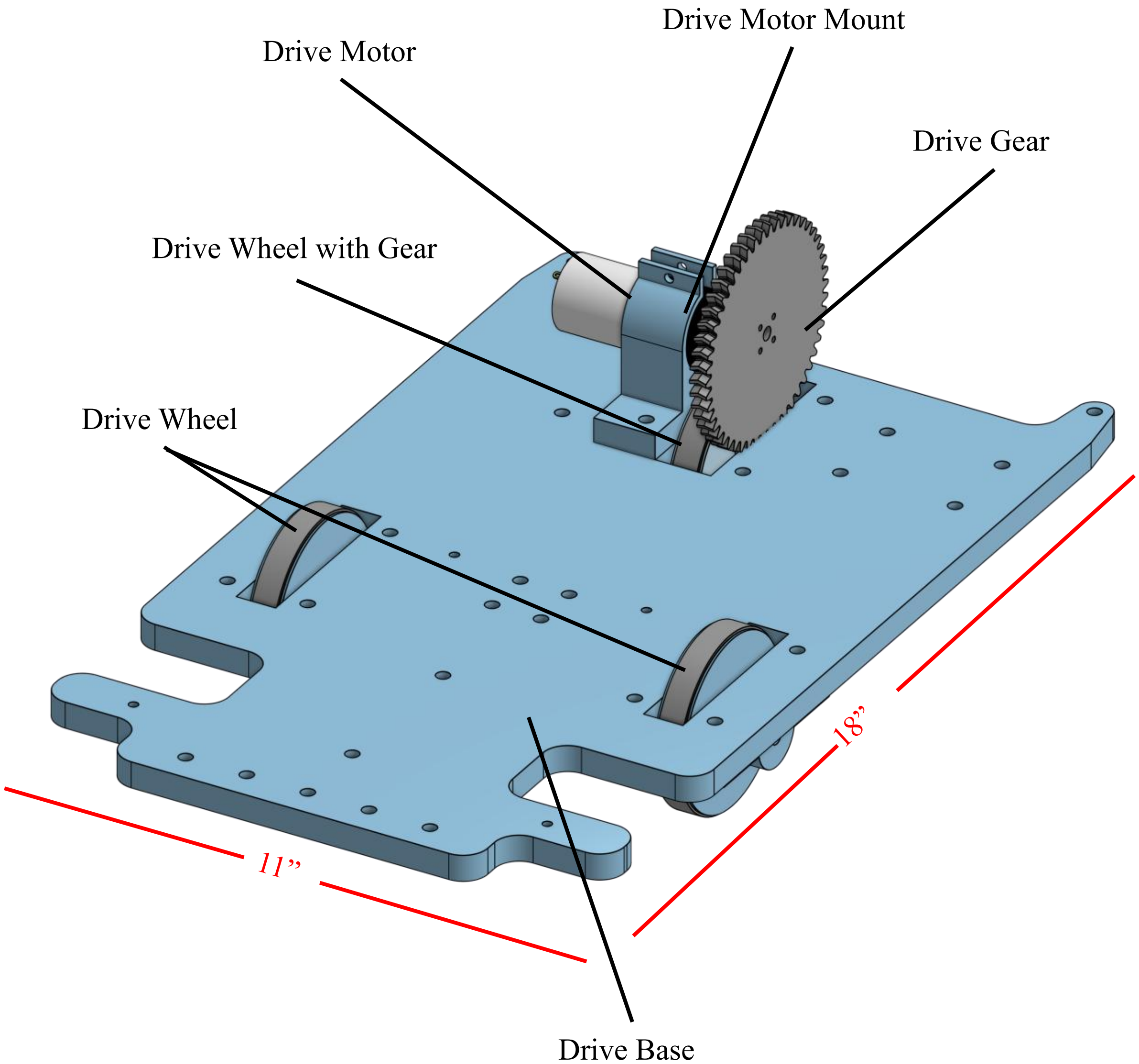


Figure 6: Drivetrain Subsystem

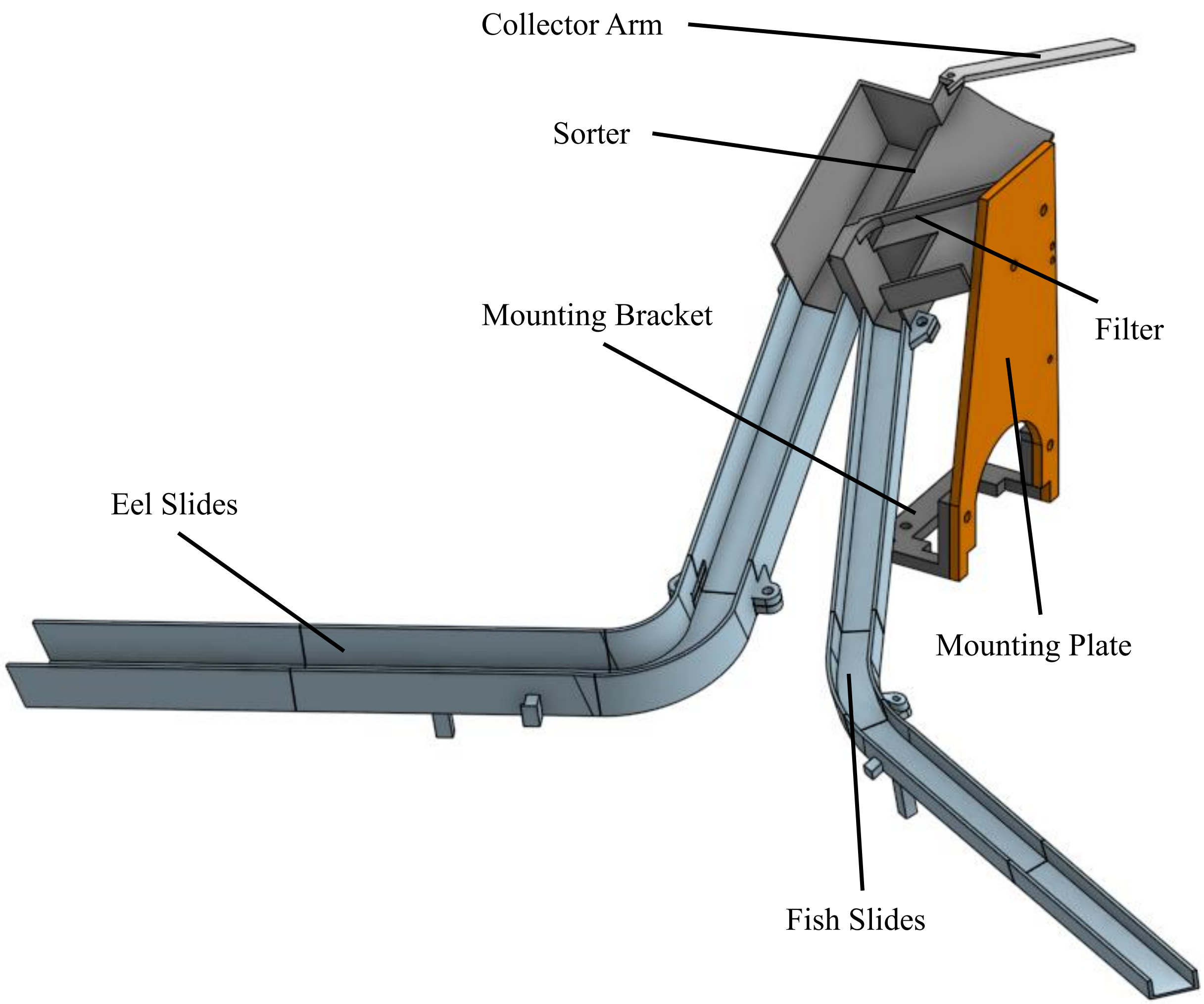


Figure 7: Fish and Eel Subsystem

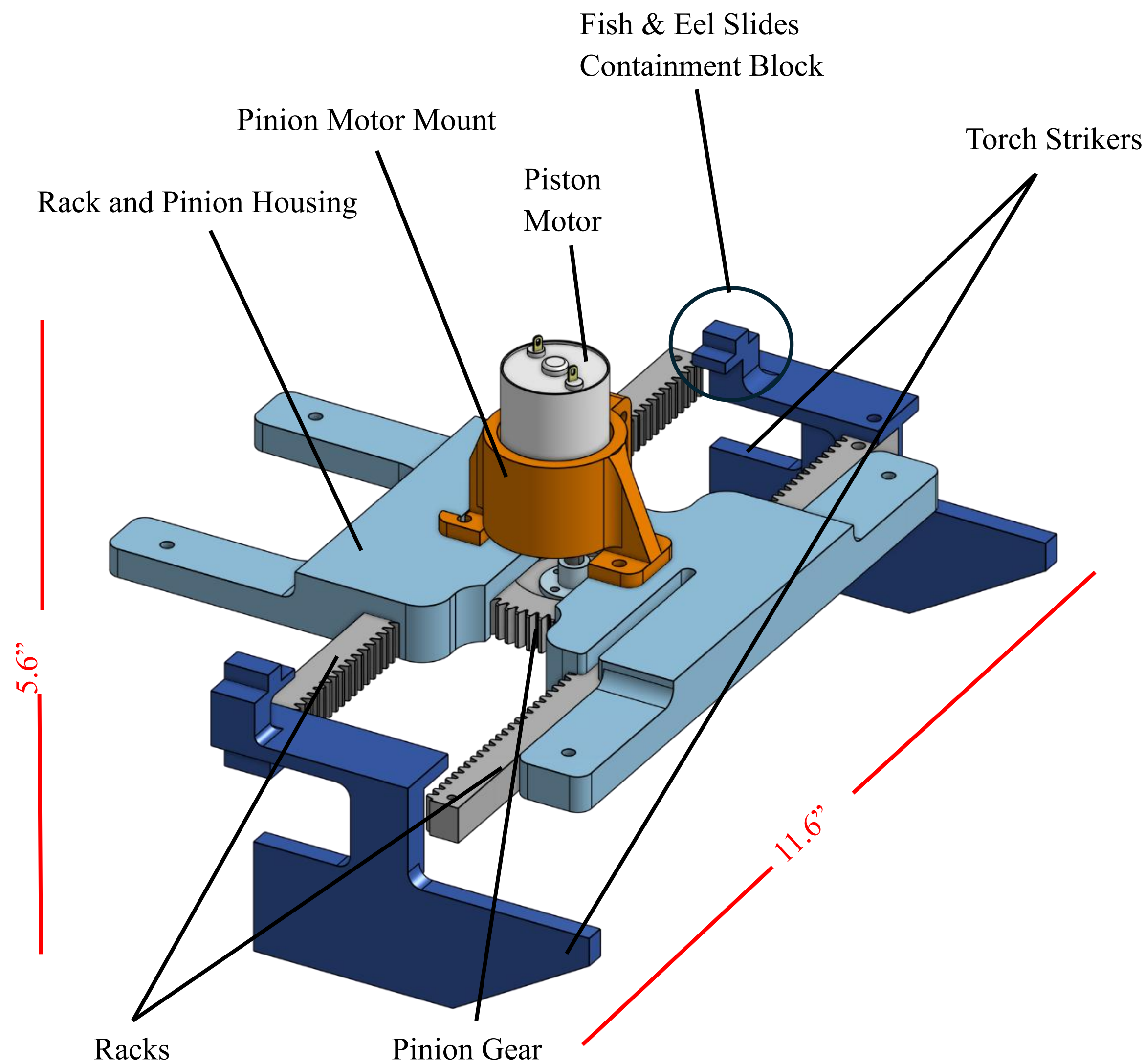


Figure 8: Torch Subsystem

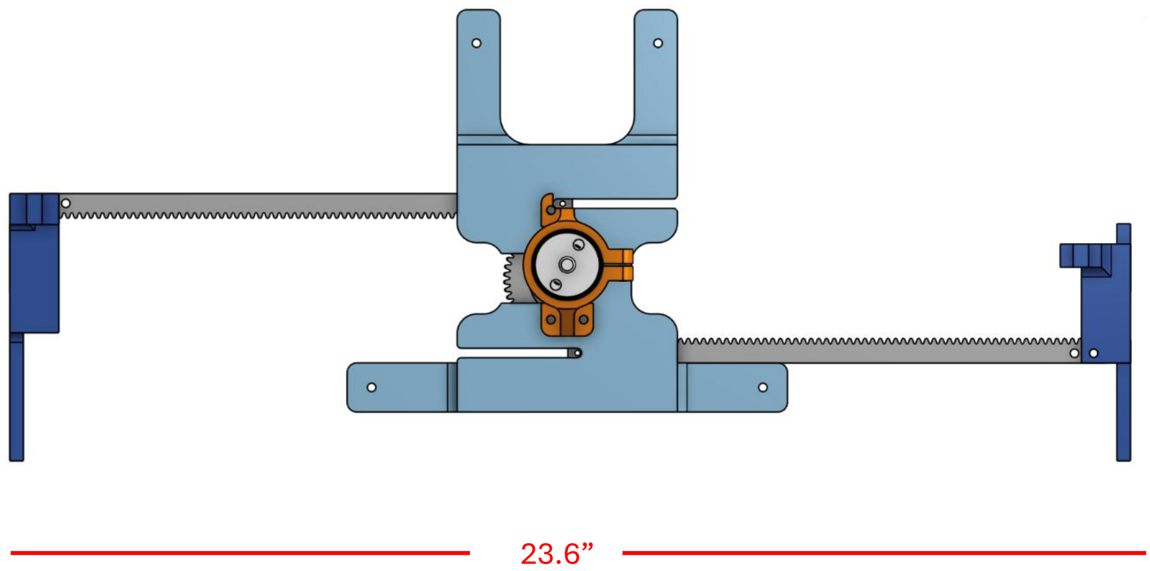


Figure 9: Torch Subsystem Top Down View

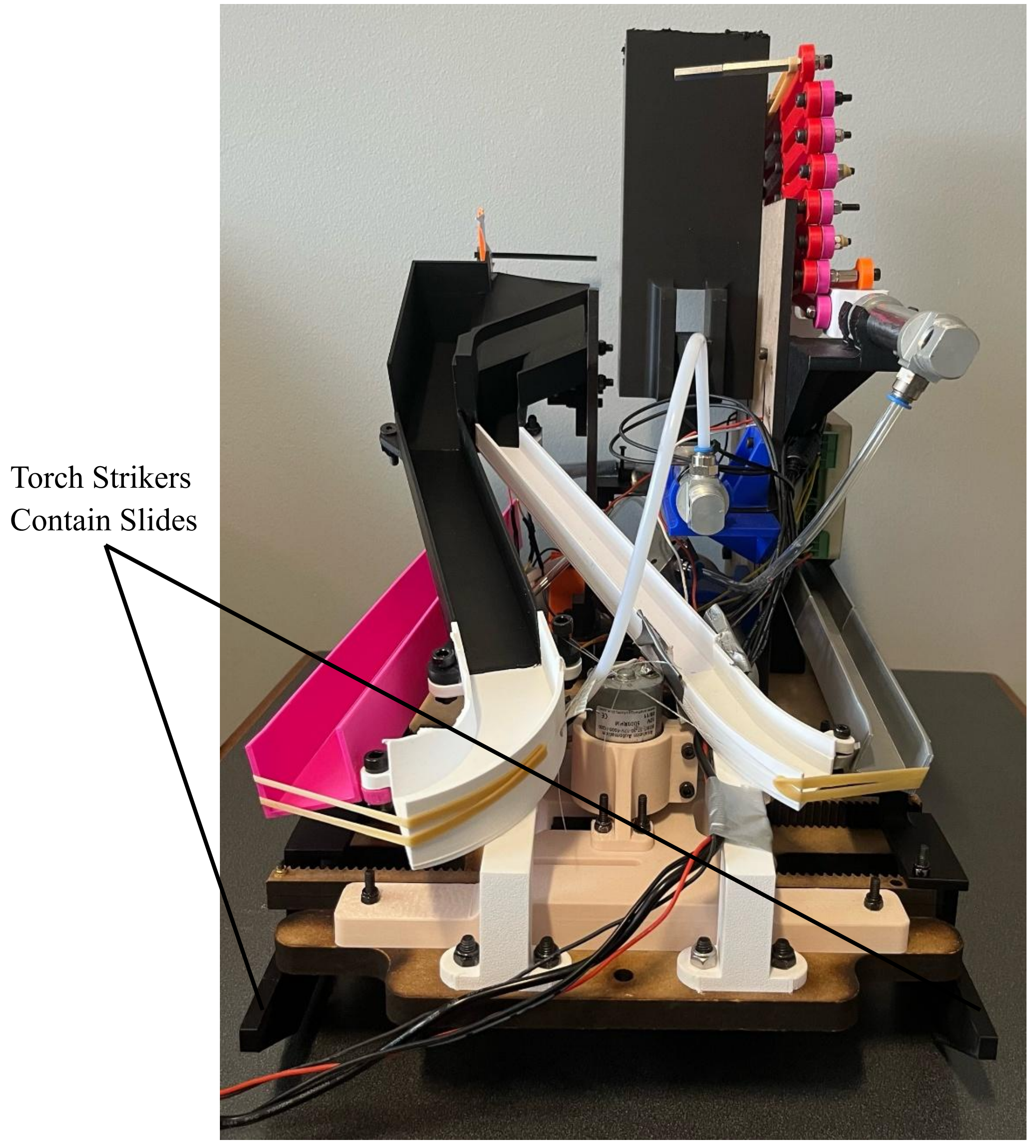


Figure 10: Robot Back View

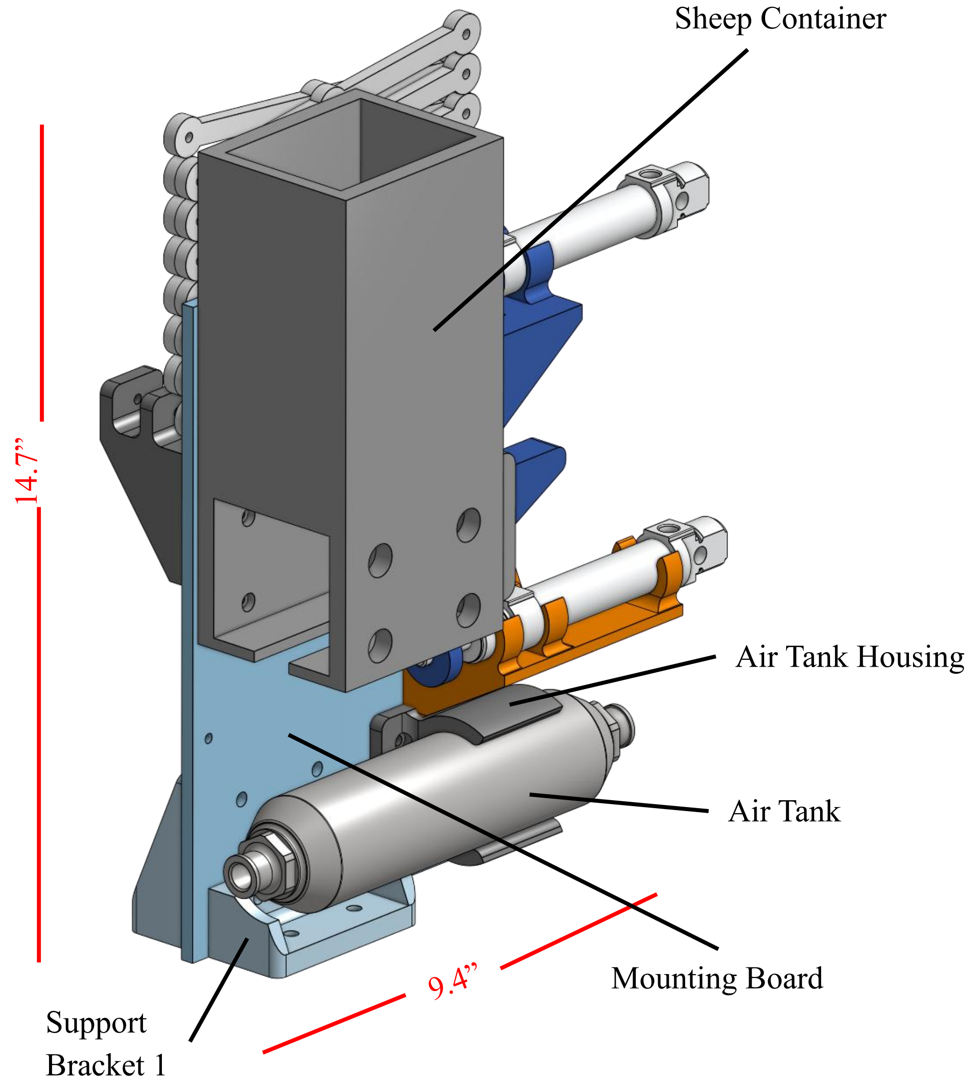


Figure 11: Sheep Subsystem

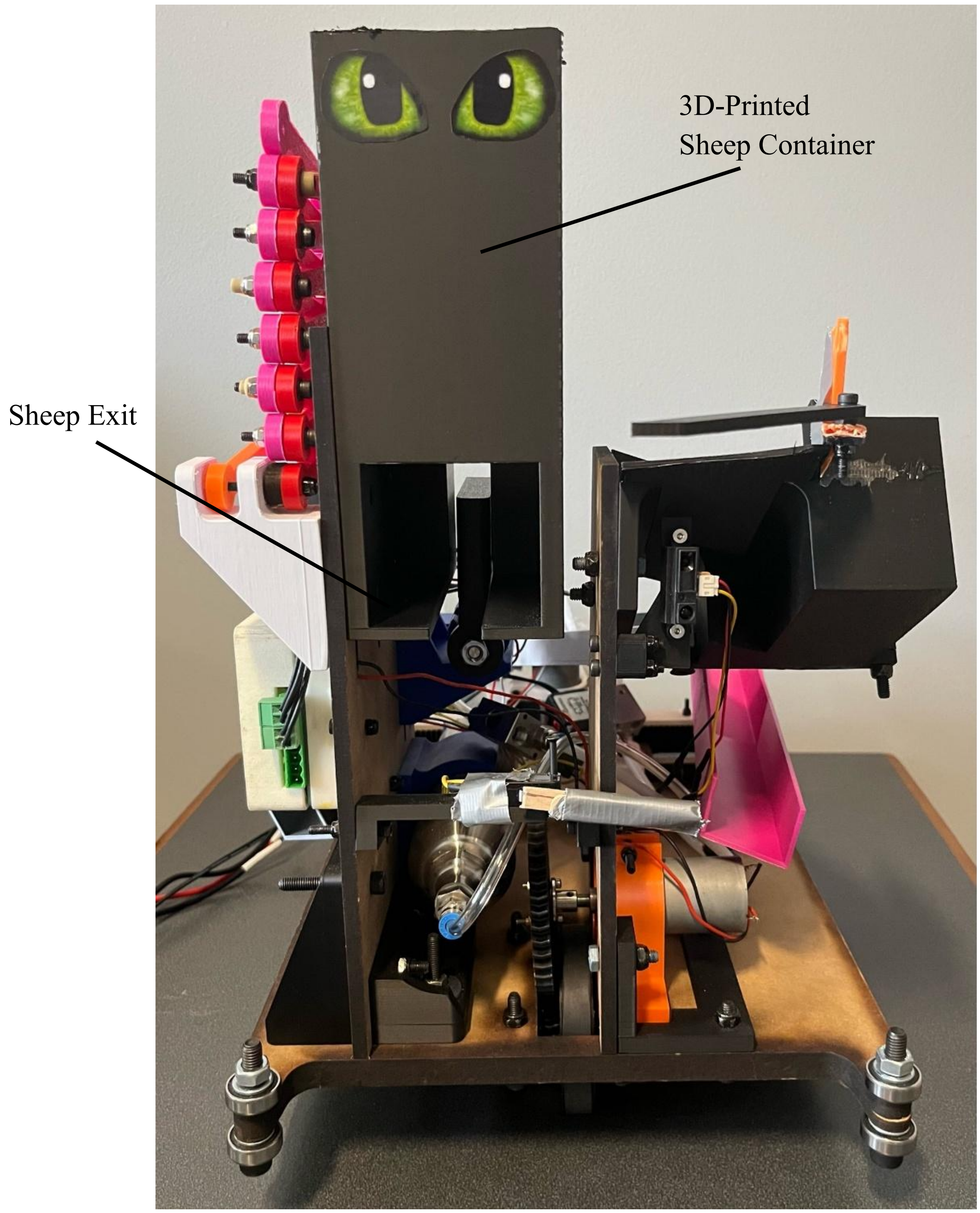


Figure 12: Robot Front View

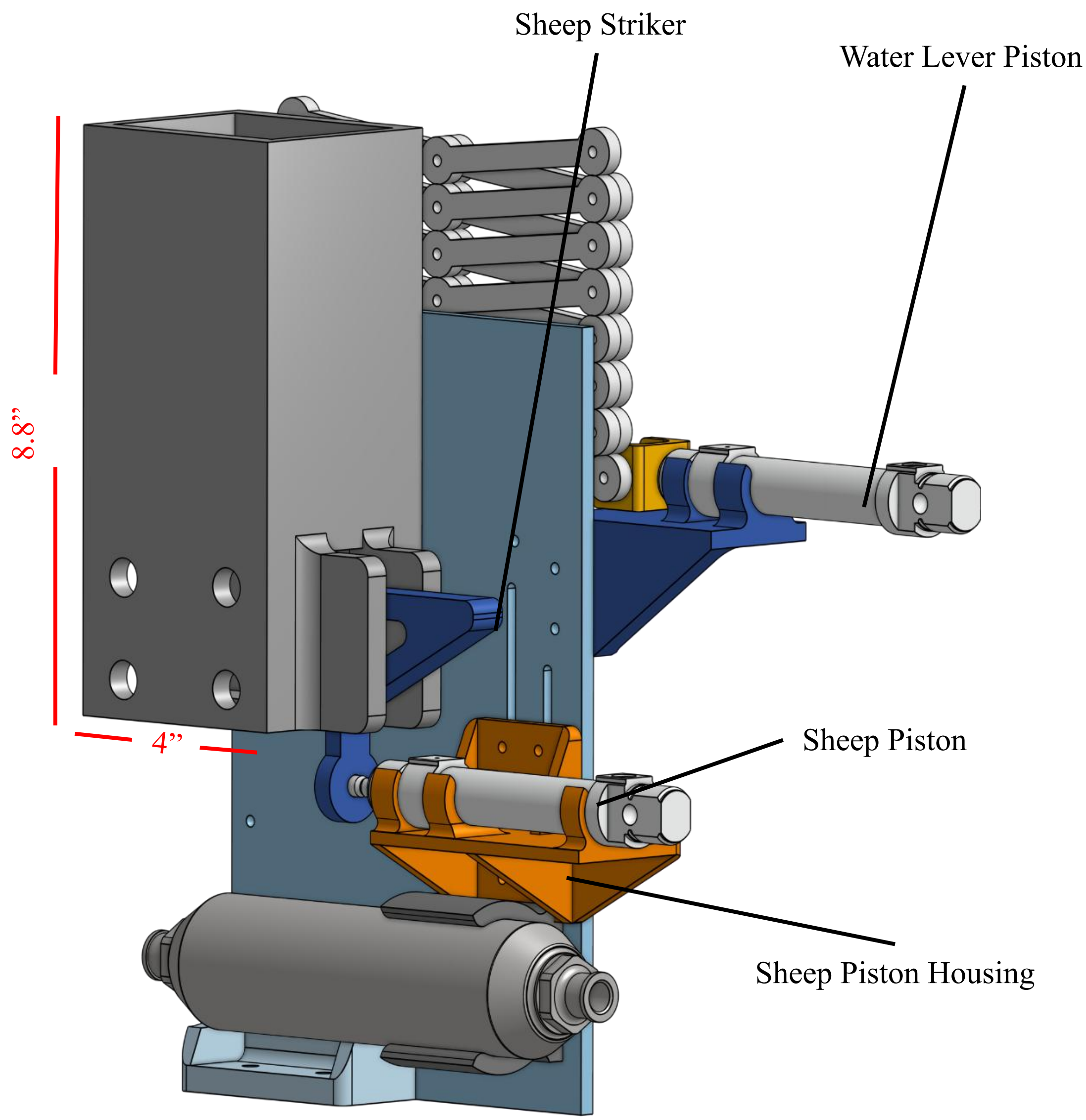


Figure 13: Sheep Subsystem Alternate View

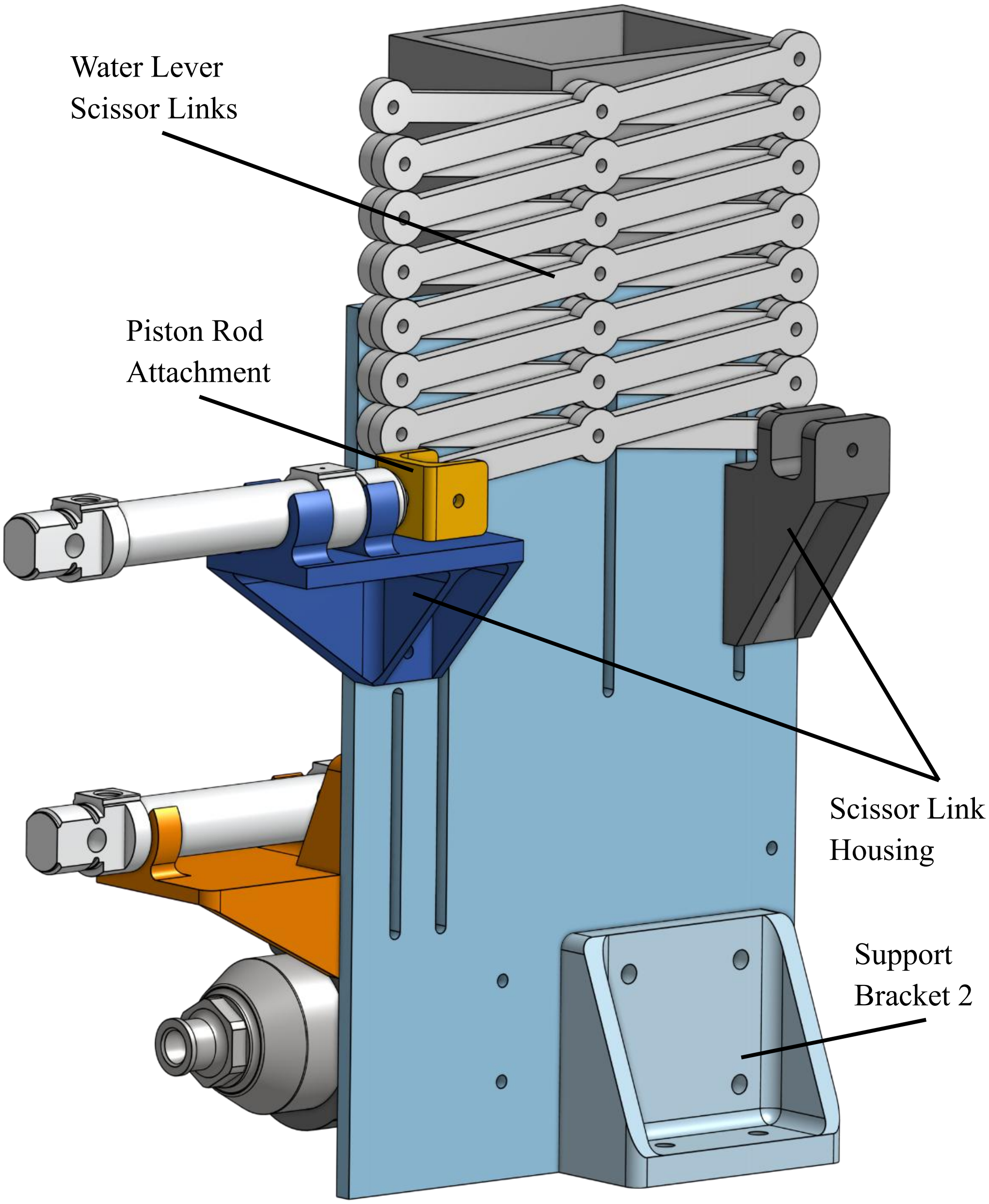


Figure 14: Water Lever Subsystem

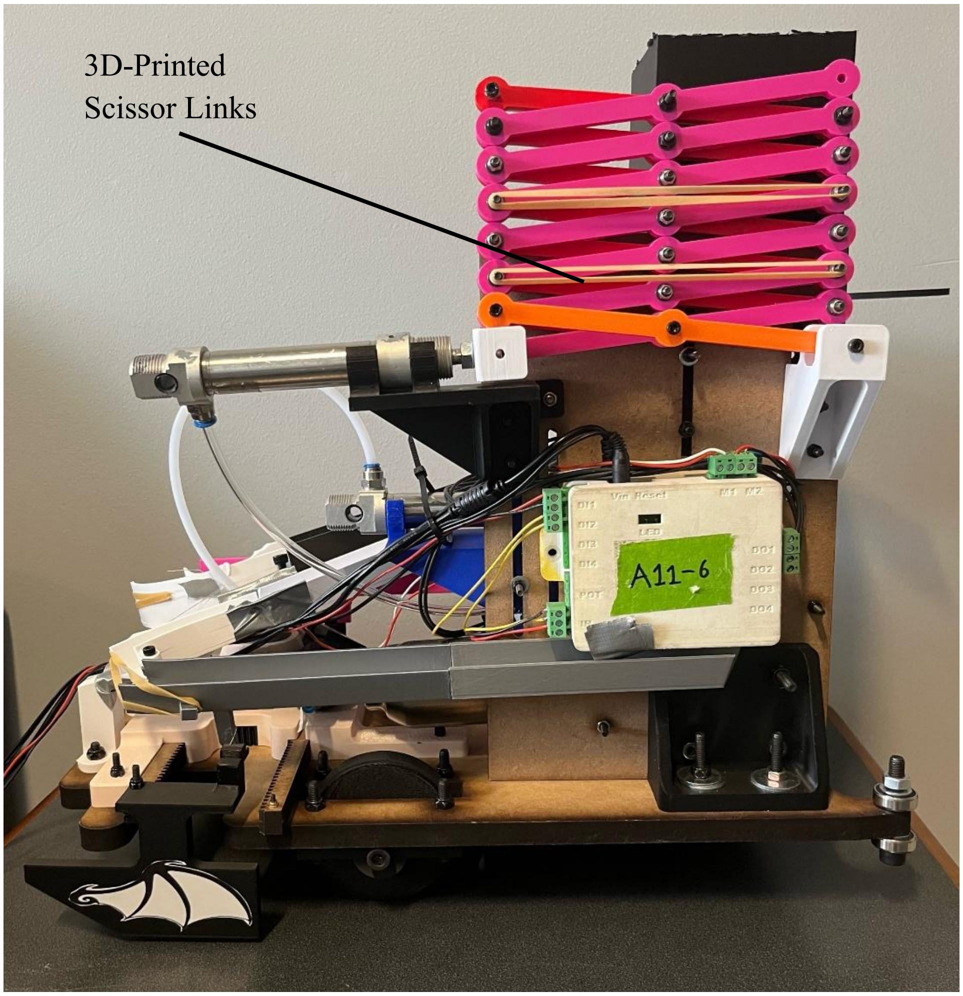


Figure 15: Robot Side View of Water Lever Subsystem

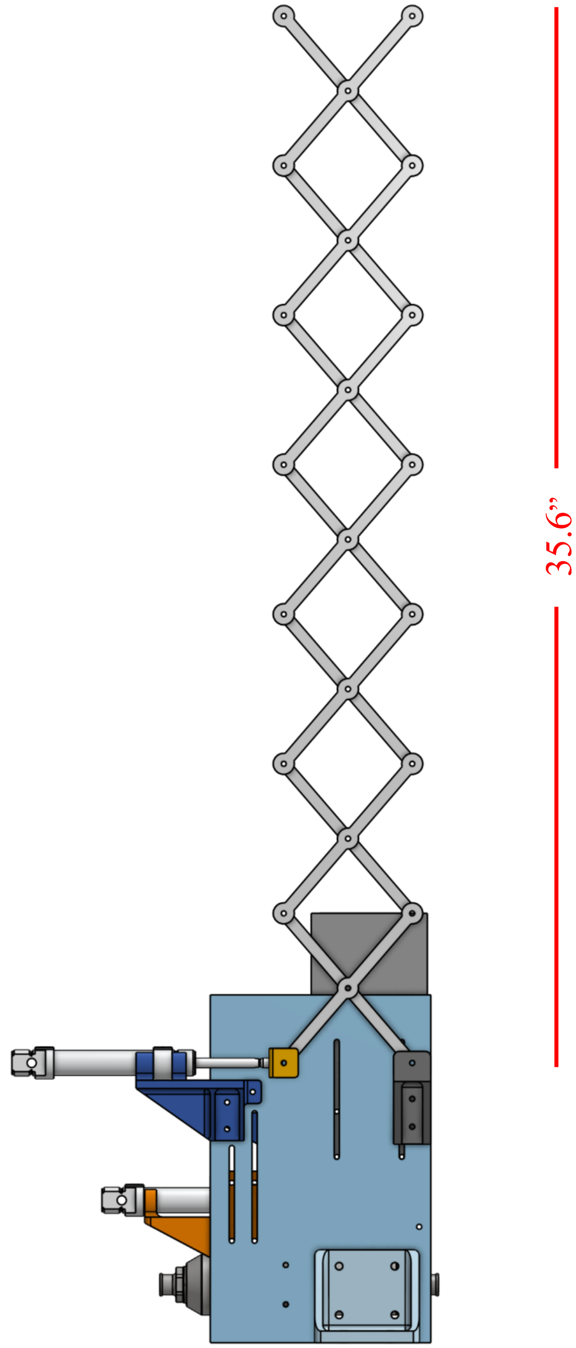


Figure 16: Water Lever Full Extension

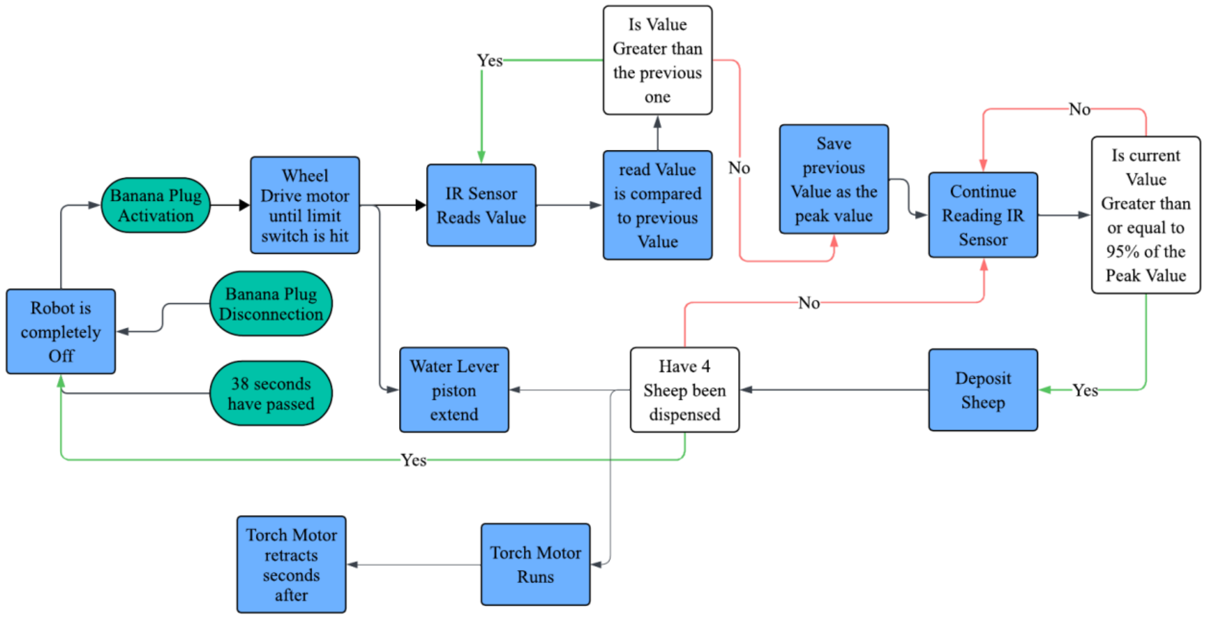


Figure 17: Robot Algorithm

Bill of Materials: Project/Product Development						
Project:	***Final Report BOM***					
Engineering Team:	How to Train Your Robot			*Megan Cahall* *Joy D'Souza* *Carlos Balbuena Munoz* *Nikhil Vemuri*		
Date:				*11/19/2025*		
						Functional Analysis
Module/ Part #	**Description/Name**	**Qty**	**Unit Cost**	**Total Cost**	**Supplier**	**Function**
1	2'x4'x1/4" MDF	0.75	$14.99	$11.24	Invention Studio	Fabrication
2	2'x4'x1/2" MDF	0.25	$14.99	$3.75	Invention Studio	Fabrication
3	1g PLA	927	$0.05	$46.35	IDEA Lab	Fabrication
4	M4x25mm Socket Head Screw	15	$0.06	$0.90	Amazon	Fastener
5	M4x30mm Socket Head Screw	15	$0.06	$0.90	Amazon	Fastener
6	M4x35mm Socket Head Screw	15	$0.06	$0.90	Amazon	Fastener
7	M4x40mm Socket Head Screw	13	$0.06	$0.78	Amazon	Fastener
8	M4x45mm Socket Head Screw	10	$0.06	$0.60	Amazon	Fastener
9	M4x50mm Socket Head Screw	10	$0.06	$0.60	Amazon	Fastener
10	M4x12mm Socket Head Cap Bolts	12	$0.30	$3.60	Amazon	Fastener
11	M4x16mm Socket Head Cap Bolt	15	$0.30	$4.50	Amazon	Fastener
12	M4x20mm Socket Head Cap Bolt	10	$0.30	$3.00	Amazon	Fastener
13	M6x20mm Socket Head Cap Bolt	10	$0.09	$0.90	Amazon	Fastener
14	M6x30mm Socket Head Cap Bolt	5	$0.09	$0.45	Amazon	Fastener
15	M8x60mm Socket Head Cap Bolt	3	$0.20	$0.60	Amazon	Fastener
16	M4 Washer	115	$0.03	$3.45	Amazon	Fastener
17	M6 Washer	15	$0.03	$0.45	Amazon	Fastener
18	M4 Nut	37	$0.04	$1.48	Amazon	Fastener
19	M6 Nut	5	$0.04	$0.20	Amazon	Fastener
20	M4x0.7mm Stainless Steel Locknut	78	$0.06	$4.68	Amazon	Fastener
21	M6x1.0mm Stainless Steel Locknut	10	$0.06	$0.60	Amazon	Fastener
22	M8x1.25mm Stainless Steel Locknut	3	$0.06	$0.19	Amazon	Fastener
23	M3x40mm Hex Standoff Spacer	1	$0.07	$0.07	Amazon	Spacer
24	M3x25+6mm Female Hex Standoff Spacer	1	$0.07	$0.07	Amazon	Spacer
25	6mm Flange Coupling Connector	2	$1.40	$2.80	Amazon	Fastener
26	1" of Adhesive Solid Rubber Strip	30	$0.05	$1.50	Amazon	Wheels
27	Steel 608 Ball Bearing	4	$0.35	$1.40	Amazon	Drive Base
28	8x14x14mm Flanged Ball Bearing	6	$1.29	$7.74	Amazon	Wheels
Total	-	-	-	**$103.70**	-	-

Figure 18: Final BOM

Item	Part Name	QTY
Drivetrain Subsystem		
1	Drive Base	1
2	Drive Wheel	3
3	Gear with D-Shaft Insert	1
4	Drive Motor	1
5	Drive Motor Mount	1
6	Drive Gear	1
Fish and Eel Subsystem		
7	Collector Arm	1
8	Sorter	1
9	Mounting Bracket	1
10	Mounting Plate	1
11	Eel Intermediate Ramp	3
12	Eel Delivery Ramp	1
13	Fish Imtermediate Ramp	3
14	Fish Delivery Ramp	1
Sheep Subsystem		
15	Sheep Container	1
16	Air Tank Housing	1
17	Air Tank	1
18	Sheep Piston	1
19	Sheep Piston Housing	1
20	Sheep Striker	1
21	Mounting Board	1
22	Support Bracket 1	1
Water Lever Subsystem		
23	Support Bracket 2	1
24	Scissor Link Housing	2
25	Water Lever Piston	1
26	Piston Rod Attachment	1
27	Scissor Link	14
Torch Subsystem		
28	Rack and Pinion Housing	1
29	Piston Motor	1
30	Piston Motor Mount	1
31	Torch Striker	2
32	Pinion Gear	1
33	Rack	2

Miscellaneous		
35	M4x25mm Socket Head Screw	15
36	M4x30mm Socket Head Screw	15
37	M4x35mm Socket Head Screw	15
38	M4x40mm Socket Head Screw	13
39	M4x45mm Socket Head Screw	10
40	M4x50mm Socket Head Screw	10
41	M4x12mm Socket Head Cap Bolts	12
42	M4x16mm Socket Head Cap Bolt	15
43	M4x20mm Socket Head Cap Bolt	10
44	M6x20mm Socket Head Cap Bolt	10
45	M6x30mm Socket Head Cap Bolt	5
46	M8x60mm Socket Head Cap Bolt	3
47	M4 Washer	115
48	M6 Washer	15
49	M4 Nut	37
50	M6 Nut	5
51	M4x0.7mm Stainless Steel Locknut	78
52	M6x1.0mm Stainless Steel Locknut	10
53	M8x1.25mm Stainless Steel Locknut	3
54	M3x40mm Hex Standoff Spacer	1
55	M3x25+6mm Female Hex Standoff Spacer	1
56	6mm Flange Coupling Connector	2
57	1" of Adhesive Solid Rubber Strip	30
58	Steel 608 Ball Bearing	4
59	8x14x14mm Flanged Ball Bearing	6
60	#5 Rubber Band	4
Total	-	**499**

Figure 19: Final Components List

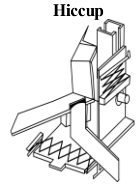
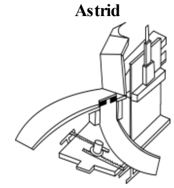
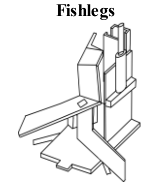
Criteria	Importance	Hiccup 		Astrid 		Fishlegs 	
		Rating	Weighted Total	Rating	Weighted Total	Rating	Weighted Total
Performs consistently	8	3	24	2	16	1	8
Able to perform in multiple rounds	9	2	18	2	18	2	18
Inactive before and after each round	10	4	40	4	40	4	40
Easy to maintain and alter	7	3	21	2	14	2	14
Professional appearance/aesthetics	2	2	4	2	4	2	4
Comply with competition rules	10	3	30	2	20	2	20
Successfully deploy	10	4	40	4	40	4	40
Fully leave dragon stables	10	4	40	4	40	4	40
Activate lever	9	4	36	1	9	3	27
Deliver fish to home zone	5	2	10	2	10	2	10
Deliver fish to feeding trough	9	3	27	2	18	1	9
Deliver eels to disposal bin	7	3	21	2	14	2	14
Deposit every sheep into unique paddock	9	3	27	2	18	4	36
Move both torches to innermost ring of bonfire	7	3	21	3	21	2	14
Total		359		282		294	
Relative Total		0.801339286		0.629464286		0.65625	
Rank		1		3		2	

Figure 20: Concept Evaluation Matrix

Table 2: Design Review and Final Competition

Design Review							Total Score
Category	Design Ingenuity	Design Implementation	Fabrication Approach	Creativity	System Explanation	Presentation Effectiveness	
Average Score	3.75	3.50	3.00	3.50	4.00	3.50	3.54
Comments	Very well designed and built modularly in case things break. Organized the team very well and combined subsystems efficiently. Very unique slide design to ensure the pieces roll well.						
	Excellent functional decoupling. They did iterations on subsystems and experimentally validated it. Good engineering team. Everyone enthusiastic and was contributing.						

Final Competition					Total Score
Heat	Time	Track	Corner Color	Final Score	
5	7:34 PM	1	White	97	180
9	8:20 PM	3	Blue	83	

References

[1] “ME2110 Fall 2025 Final Project 5.1.” Atlanta: George W. Woodruff School of Mechanical Engineering, 2025.

Contributions Statement

1. Megan Cahall — Contributed toward Figures,
2. Joy D’Souza — Contributed toward Figures, Abstract, Components list,
3. Carlos Balbuena Munoz — Contributed toward Figures,
4. Nikhil Vemuri — Contributed toward Figures